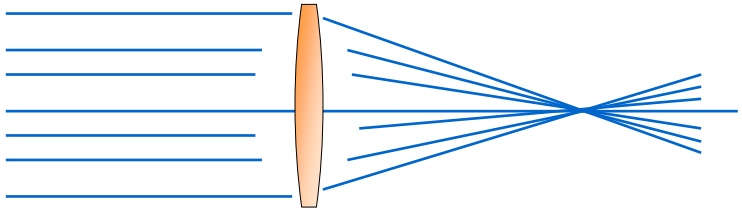
*PHYC10009 Foundations of Physics*

# Intensity, Interference

# 14



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## 12.2 Power, intensity

- *Power* is energy per unit time
- *Intensity* = *power per unit area*

$$I = \frac{\text{Power}}{\text{Area}} = \frac{P}{A}$$

- Recall from SHM, energy is $E = \frac{1}{2}kx_m^2$ ($k$ is spring constant)
  $x_m$ is amplitude (i.e. max $x$): $x(t) = A\cos(\omega t) \equiv x_m \cos(\omega t)$

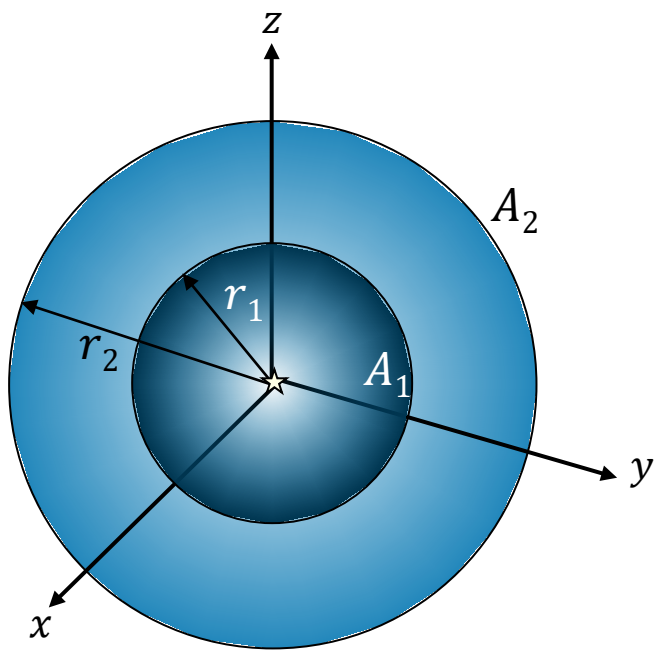
- Intensity is proportional to rate at which energy is transferred

- Infer $I \propto x_m^2$ where proportionality depends on the type of wave

The intensity of a wave is proportional to the square of its amplitude

# *Inverse square law*

- Intensity is power per unit area $I = \frac{\text{Power}}{\text{Area}}$
- Area of sphere $A = 4\pi r^2$




$$I = \frac{\text{Power}}{\text{Area}} = \frac{P}{4\pi r^2} \quad \text{(sphere)}$$

# *Intensity and sound*

- Sound waves: $s(x,t) = s_m \cos(kx - \omega t)$

  … (a few steps) …

- We find: $I = \frac{1}{2}\rho v \omega^2 s_m^2$

  – $\rho$ = density, $v$ = speed, $\omega$ = frequency, $s_m$ = molecular displacement

- Measure sound intensity → molecular displacement!

- Example: coming soon. First, we look at how to measure sound energy.

# 12.2 Power, intensity, decibels

- Intensity from displacement amplitude, $I \propto s_m^2$:

$$I = \frac{1}{2}\rho v \omega^2 s_m^2$$

- *Power* $(P = I\ A)$ varies over <u>many</u> orders of magnitude

- E.g. Human hearing:
  - Threshold 1 x $10^{-12}$ W/m$^2$
  - Pain threshold 10 W/m$^2$
  - Thirteen (13) orders of magnitude! 1 cent vs. \$100,000,000,000 (\$100 billion)

Consider 100W light at 3m: ~1W/m$^2$

*Logarithmic scale: decibels* $\beta_{\mathrm{dB}} = 10 \log_{10}\left(\frac{I}{I_0}\right)$

where $I_0$ is reference level; choose threshold value, i.e. 1 x $10^{-12}$ W/m$^2$

# 12.2 Power, intensity, decibels

*Logarithmic scale: decibels* $\beta_{\mathrm{dB}} = 10\log_{10}\left(\frac{I}{I_0}\right)$

where $I_o = 1 \times 10^{-12}$ W/m$^2$

- Thus at threshold $\beta_{\mathrm{dB}} = 10\log_{10}\left(\frac{10^{-12}}{10^{-12}}\right) = 10\log_{10} 1 = 0\ \mathrm{dB}$
- And at pain threshold $\beta_{\mathrm{dB}} = 10\log_{10}\left(\frac{10}{10^{-12}}\right) = 10\log_{10} 10^{13} = 10 \times 13 = 130\ \mathrm{dB}$
- Add 10dB for every *factor* of 10 in intensity

# *Concept Test*

At dinner time, my daughter is ten times louder than her brother (otherwise she doesn't get a word in). The sound level of my son's voice is 60dB, in which case my daughter's is

**a) 61 dB**
**b) 70 dB**
**c) 100 dB**
**d) 600 dB**

# *Typical sound levels*

| Source | Sound level dB(A) | Description |
|---|---|---|
| Military jet, Formula 1 at 15m | 130 | Painfully loud |
| Jet helicopter at 30m | 100 | Very loud |
| Power mower | 96 | Annoying |
| Living room music | 76 | Depends on the music |
| Conversation at 1m | 60 | |
| Soft whisper at 2m | 35 | |

# *Sample exam question: 2017*

**Question 6** **6 marks**

You are driving a massive truck and become stuck at a railway crossing. Your truck completely blocks the railway tracks, which extend in a straight line in both directions. You hear a train approaching when it toots its horn. Let $I_1$ be the intensity of the sound you hear from the horn.

(a) In terms of $I_1$, what is the intensity of the horn, $I_2$, when the train is closer, at half the original distance?

(b) If $I_1 = 70$ dB, calculate the intensity $I_2$ in dB.

**[3 + 3 = 6 marks]**

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$$\mathrm{I}(r) = \frac{I_0}{r^2}$$

(a) Let $I_1 = \frac{I}{r^2}$, Intensity at Distance r, $I_2 = \frac{I}{(\frac{r}{2})^2} = \frac{4I}{r^2} = 4I_1$

(b) $\beta_{\mathrm{dB}}(I_2) = 10log_{10}(I_2) = 10log_{10}(4I_1) = 10log_{10}(4)$+ $10log_{10}(I_1)$=6+70 = 76 dB

# *Example exam question*

For a sound wave, the intensity is $I = \frac{1}{2}\rho v \omega^2 s_m^2$ where $\rho$ is density, $v$ is velocity, $\omega$ is frequency and $s_m$ is displacement $s_m$.
Use air density 1.225 kg/m$^3$, sound velocity 343 m/s, frequency 440 Hz

a) Calculate typical displacement of air molecules at threshold of hearing, i.e. intensity of $10^{-12}$ W/m$^2$.

b) Calculate $s_m$ for intensity 120 dB.

# *Superposition*

- If we add two waves, the amplitude of the combined effect is just the sum of the individual waves

- If $D_1, D_2$ are two allowed waves then so is $D = D_1 + D_2$

- *At a given point, overlapping waves add algebraically, to produce a resultant wave*

- *The travel of a wave is unaffected by interaction with other waves*

http://mysite.verizon.net/vzeoacw1/wave_interference.html

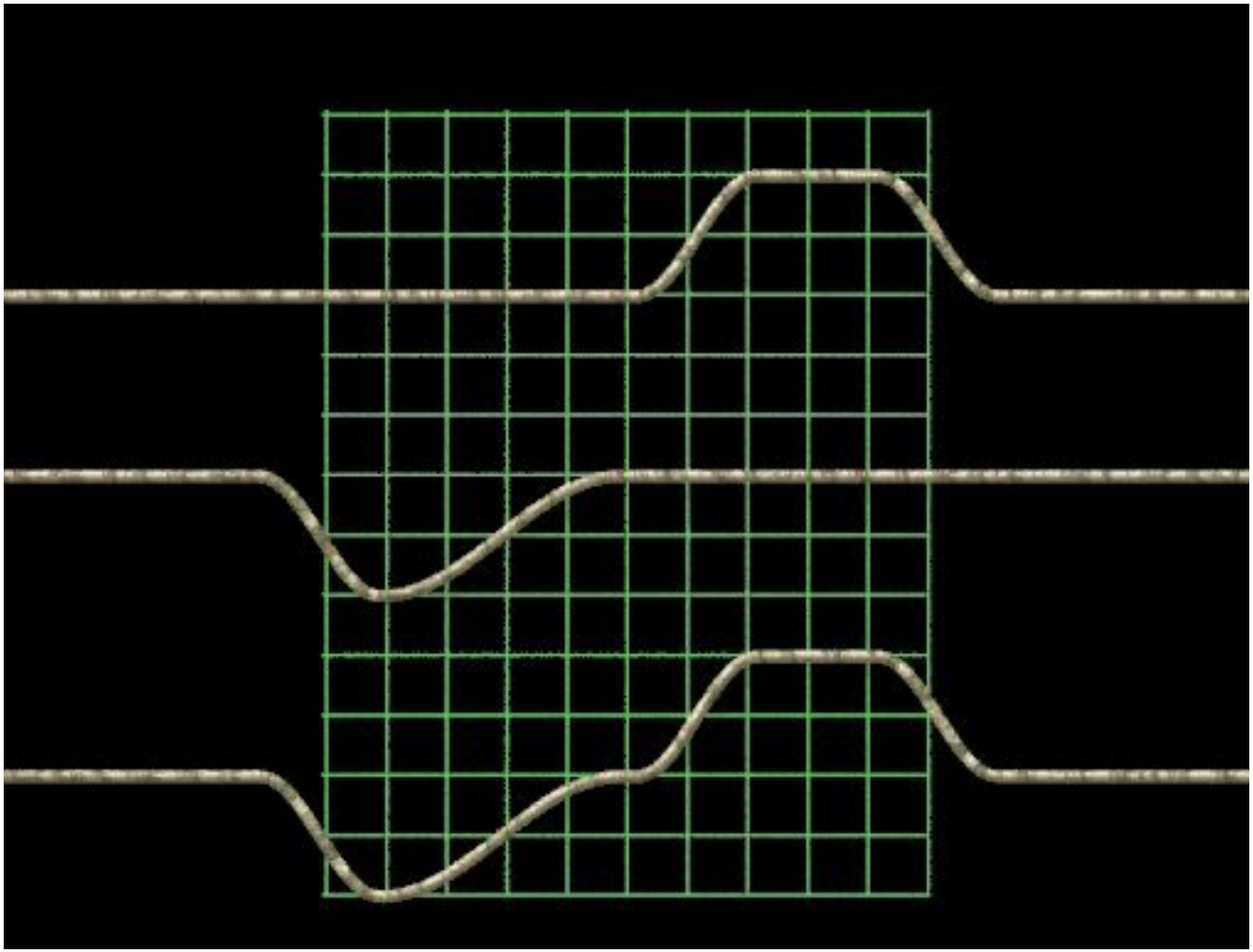
## *The principle of superposition*

- When two or more waves are *simultaneously* present at a single point in space, the displacement of the medium at that point is the sum of the displacements due to each individual wave

$$D(x,t) = \sum_i D_i(x,t)$$

# *Principle of Super Position*

**Pulses interfering**



# *Making (standing) Waves*

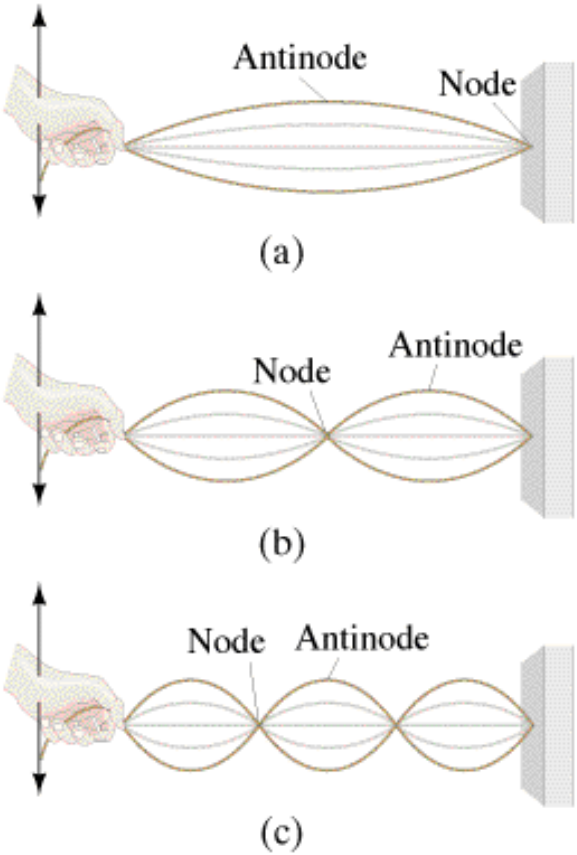
When a wave is confined it can reflect from the confining wall

# *Interference: Standing waves (16.1/16.2)*

If the reflection conserves the wave energy, and the size of the container forms *nodes* at the ends of cavity, a ***Standing Wave*** is formed

- Oscillate string with a fixed termination – wave propagation
- Interference of waves traveling back and forth along the string
- At appropriate frequencies, the two waves interfere to form a large-amplitude **standing wave**
- An example of

**Resonance**

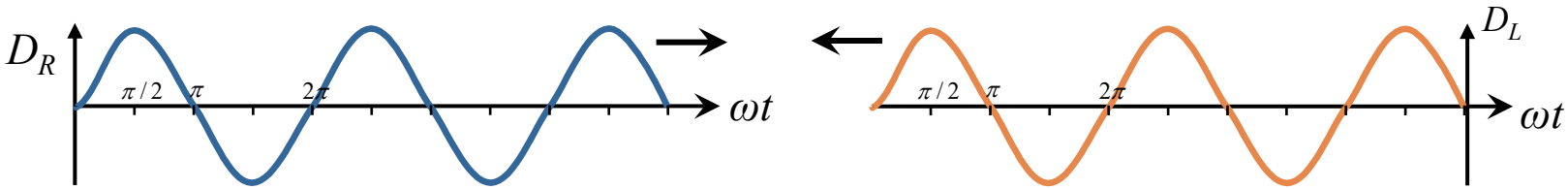


# *Superposition: standing waves*

- Superposition (interference)
  - same frequency, wavelength, velocity, amplitude
  - *different* direction: counter-propagating

Two identical counter-propagating travelling waves

$$D_R = a\cos\left(\tfrac{2\pi}{\lambda}x - \omega t\right) \qquad D_L = a\cos\left(\tfrac{2\pi}{\lambda}x + \omega t\right)$$




$$\Rightarrow D(x,t) = D_R + D_L = \left[2a\cos\left(\tfrac{2\pi}{\lambda}x\right)\right]\cos(\omega t)$$

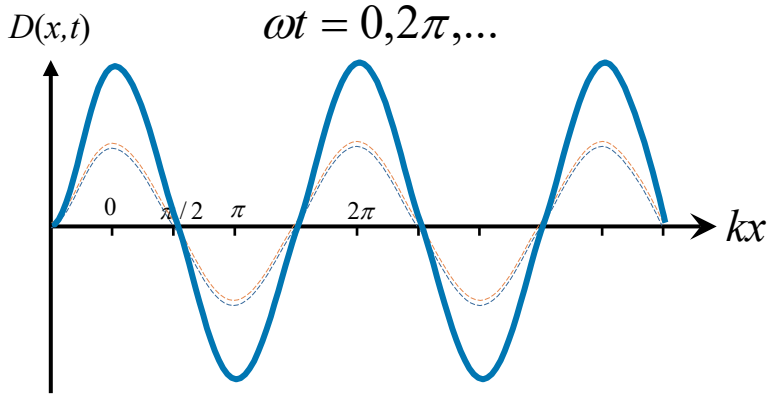
$$D(x,t) = A(x)\cos(\omega t)$$

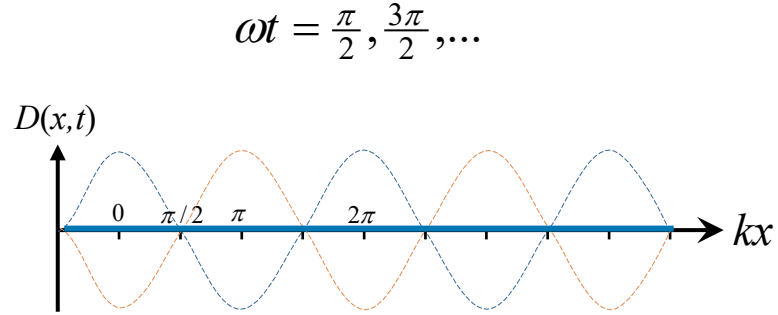
<u>Not</u> a travelling wave

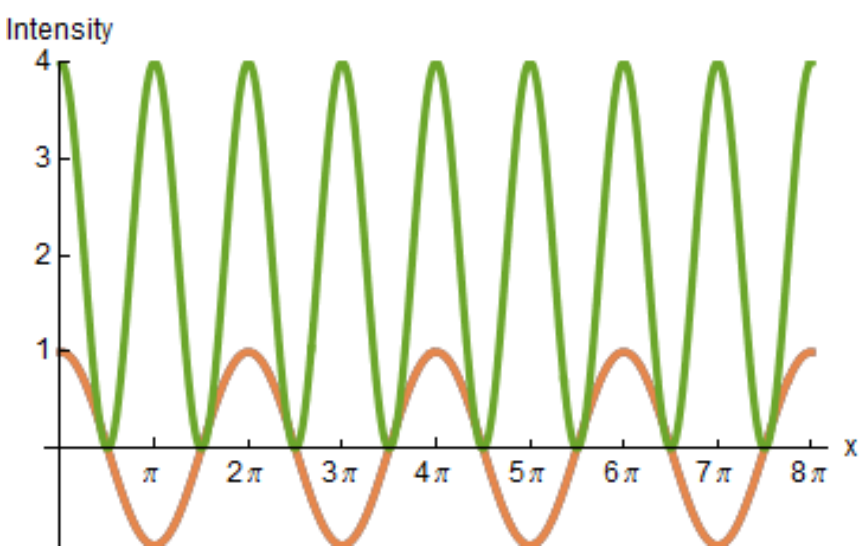
# *Superposition: standing waves*

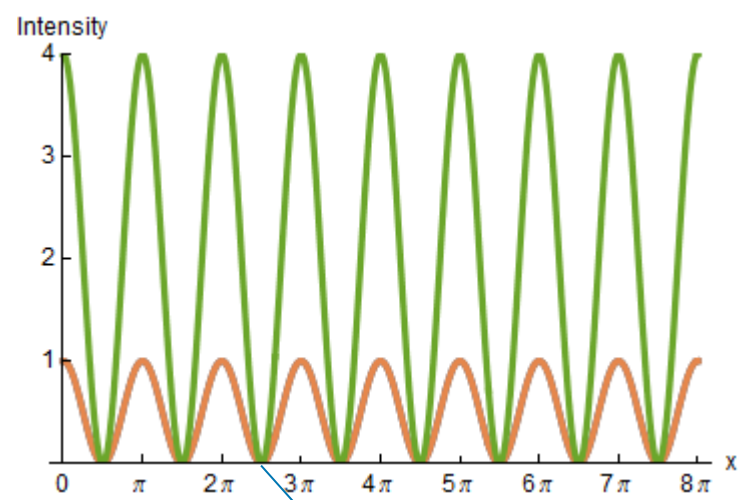
Two identical counter-propagating travelling waves

**D** **Wg-2 Beats using signal gens**



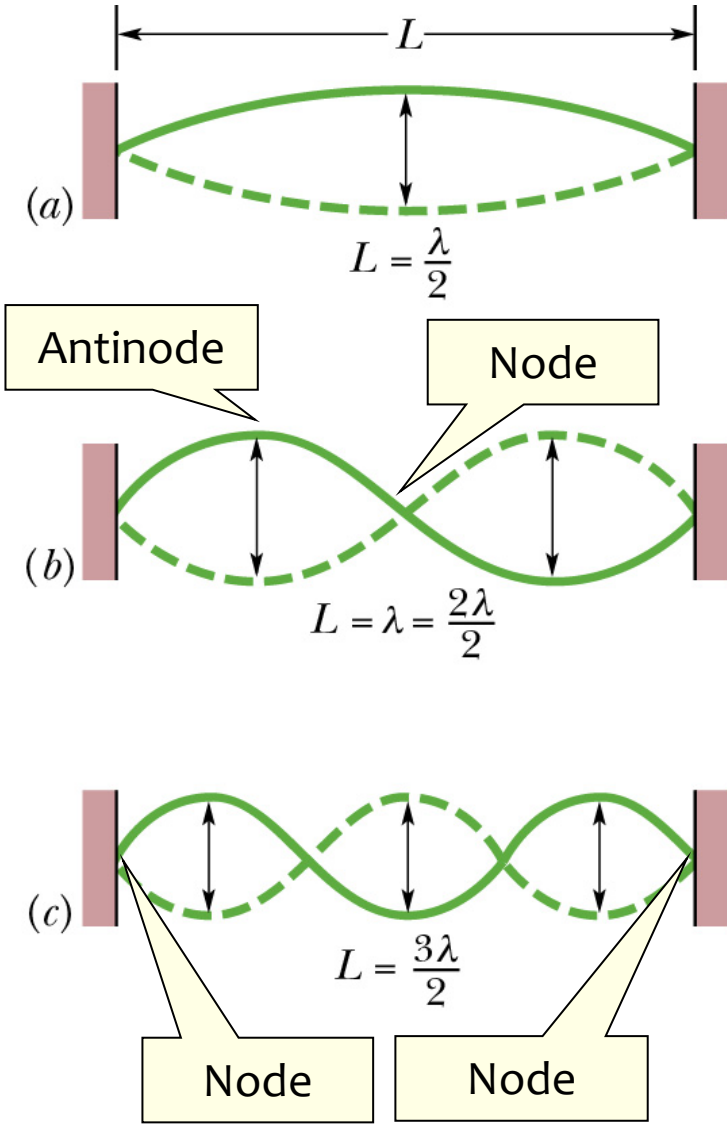




Mobile phone dead-zones!

# *Superposition: standing waves*

$$D(x,t) = \left[2a\sin\left(\tfrac{2\pi}{\lambda}x\right)\right]\cos(\omega t)$$

- Boundary conditions *limit the allowed wavelengths*
- *MUST have nodes at each end!*




- Frequencies

$$f_m = m\frac{v}{2L} = mf_1$$

- Lowest frequency $f_1$ is the *fundamental*

  *$m$ is the "mode number": the number of antinodes*

- Separation of nodes $\lambda/2$

This is a simple example of resonance: frequencies allowed are those with integer half-wavelengths. Important for electron wavefunction concepts later

D **Wb-1 Melde' Experiment**

**D** **Wd-4 Smash Wine Glass**